INCH-POUND

A-A-59994
December 4, 2018

COMMERCIAL ITEM DESCRIPTION

BUTTON, SEWING HOLE, AND BUTTON, STAPLE, (PLASTIC)

The General Services Administration has authorized the use of this commercial item description (CID) as a replacement for V-B-871 for all federal agencies.

1. SCOPE

1.1 Scope. This CID covers the requirements for plastic staple button and sewing-hole button. Toggles and rings for staple buttons are also covered. The buttons are intended for use in securing clothing and equipment, and may also serve decorative purposes.

2. CLASSIFICATION

2.1 Classification. The buttons shall be available in the following types, styles and classes, as specified. Buttons styles are shown in Figure 1 (Type I) and Figure 2 (Type II). Tables IV and V outline historical changes and deletions to specifications relative to types, classes and styles.

2.1.1 Types and styles. The following types and styles (see Figures 1 and 2) are currently active for use by the DoD.

Type I. Sewing-hole buttons.

Style 4
Style 15
Style 19
Style 20
Style 21
Style 24
Style 24A
Style 25
Style 26
Style 29
Style 30

Beneficial comments, recommendations, additions, deletions, clarifications, etc. and any data that may improve this document should be sent to: DLA Troop Support Standardization Team, 700 Robbins Avenue, Philadelphia, PA 19111-5096. Since contact information can change, verify the currency of the address information using Acquisition Streamlining and Standardization Information System (ASSIST) online database https://assist.dla.mil.

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Type II. Staple buttons.

Style 1
Style 3
Style 5
Style 6

2.1.2 Classes.

Class 1 - Melamine formaldehyde composition
Class 2 - Pearlescent-polyester composition

2.1.3 Size. The size of the buttons shall be specified in the contract or purchase order. Typical sizes for the styles specified in 2.1.1 and classes in 2.1.2 are listed in Table I.

3. SALIENT CHARACTERISTICS

3.1 Material.

3.1.1 Button molding compound. The button molding plastic material shall be composed of a melamine formaldehyde or pearlescent-polyester, as specified (see 2.1.2 and 7.5) of a quality which will be suitable for producing buttons conforming to the requirements specified herein.

3.1.2 Metal components (Type II staple buttons).

3.1.2.1 Metal shanks. Shanks shall be made of drawn round or half-round steel, wire, zinc alloy or brass, except that bell or undercut shanks, stab or screw shanks may be stamped from brass or zinc or die cast from zinc alloy. Shanks shall meet all applicable requirements cited herein.

3.1.2.2 Rings and toggles. Rings and toggles shall be made of hard-drawn spring steel wire conforming to commercial standards and shall conform to the requirements specified herein.

3.1.2.3 Metal finish. All metal components shall be plated with any suitable white metal or metallic compound. Cadmium shall not be used.

3.2 Color and finish.

3.2.1 Color. The button shall match the standard sample for color (see 7.4) as specified and shall be evaluated in accordance with AATCC Evaluation Procedure 9, Option A. The viewing conditions shall consist of sources simulating artificial daylight D75 illuminant with a color temperature of 7500 (±200) K illumination of 100 (±20) foot candles, under incandescent A illuminant with a color temperature of 2856 (±200) K. Unless otherwise specified, the color shall be consistent throughout the thickness of the button.

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3.2.2 Finish (end-items). Buttons shall have a glossy finish unless otherwise specified (see 7.5). The finish shall be uniform and a reasonably close match to the standard button sample (see 7.4) when examined in the viewing conditions specified in 3.2.1.

3.3 Physical requirements.

3.3.1 Dimensions. The dimensions of the buttons shall be as shown in Table I. The size and thickness of the buttons is measured in ligne using a calibrated micrometer. One (1) ligne is equal to 0.025 (1/40) inch. Tolerance shall be 1/2 ligne (±0.0125 inch) for Class 1 buttons and 1 ligne (±0.025 inch) for Class 2 buttons.

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TABLE I. Dimensions of finished buttons (typical DoD button sizes and styles).

	Styles for Class 1	Styles for Class 2	Size (diameter)		Minimum thickness 2/ 3/	
			Ligne	Inch 1/	Ligne	Inch 1/
Type I buttons	15 or 19	-	18	0.450	4	0.100
	15, 19, 24, 24A or 29	-	20	0.500	4	0.100
	24 or 24A	-	22	0.550	4	0.100
	4 or 19	-	22	0.550	5	0.125
	19, 20 or 21	-	24	0.600	5	0.125
	-	25	14	0.350	4	0.100
	-	25	16	0.400	4	0.100
	-	25	17	0.425	4	0.100
	-	25	19	0.475	4	0.100
	-	25	21	0.525	4	0.100
	4	-	27	0.675	5	0.125
	19, 20 or 21	-	30	0.750	5	0.125
	20 or 21	-	34	0.850	5	0.125
	20 or 21	-	40	1.000	7	0.175
	20 or 21	-	45	1.125	7	0.175
	26	-	30	0.750	8	0.200
	30	-	36	0.870	6	0.150
Type II buttons	-	5 4/	21	0.525	10	0.250
	-	6 5/	21	0.525	10	0.250
	-	3	22	0.550	4	0.100
	-	3	24	0.600	4	0.100
	1	-	30	0.750	8	0.200
	-	3	30	0.750	4	0.100

1/ Decimal equivalent.

2/ For Type II buttons, thickness shall be measured without staples.

3/ Thickness measurements shall be reported to the nearest full ligne. Actual thickness shall be measured and recorded but the value obtained shall be carried to the nearest full ligne for reporting purposes. For example: A measurement of ligne 3.49 is reported as a ligne 3, while a measurement of ligne 3.5 is reported as a ligne 4.

4/ Style 5 is hemispherical in shape, and the shank (including the eye) shall protrude from the back of the button not less than 0.125-inches (5 ligne) and not more than 0.188 inches (7-1/2 ligne).

5/ Style 6 is hemispherical in shape. The 12-1/2 ligne shank (including eye) shall protrude from the back of the button not less than 0.313-inches (12-1/2 ligne) and not more than 0.375-inches (15 ligne).

3.3.2 Resistance to chalking (Class 1 buttons only). Class 1 buttons shall not exhibit chalking when tested as follows: Buttons shall be immersed in a boiling solution of 0.8 percent by weight sulfuric acid for ten (10) minutes, and immediately thereafter rinsed in water, dried and

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examined under the viewing conditions specified in 3.2.1. Any chalking, as evidenced by a dry chalk-like appearance or deposit on the surface of the button, shall be considered a test failure.

3.3.3 Compressive strength. The compressive strength shall be tested as specified in either 3.3.3.1 or 3.3.3.2. Type I buttons shall meet the minimum requirements shown in Table II for the specified class and style. Type II buttons shall have a compressive strength of not less than 100 pounds, regardless of style. The results for all the buttons in the sample shall be averaged and reported to the nearest 1 pound. Failure of the average to conform to the specified requirement shall be considered a test failure.

TABLE II. Compressive strength of Type I buttons.

Class	Style	Average compressive strength - pounds (minimum)
1	4	1000
1	15	1000
1	19	500
1	20 or 21	800
1	24 or 24A	150
1	26	1800
1	29	1000
1	30	130
2	25	500

3.3.3.1 ASTM D695. Compressive strength testing of buttons may be conducted in accordance with ASTM D695. When testing staple buttons, the shank shall be removed even with the back of the buttons. In case of conflict, test equipment with an automatic stress-strain recorder shall be used for referee testing. The rate of head travel shall not exceed 0.050-inches per minute. The compressive strength shall be taken from the chart as the highest reading in pounds prior to the first drop in the curve which occurs when the button cracks.

3.3.3.2 Visual or audible cracking. Compressive strength testing of buttons may be conducted using apparatus which permits gradual application of the load either by a manual or automatic hydraulic mechanism until button cracking occurs. The load indicating mechanisms (gauges) shall be capable of showing the total compressive load carried by the button. However, the gauge shall be so calibrated that that the load deflection at the point of button failure is at least one-third of the total capacity of the gauge. Buttons shall be placed face-down, one at a time, between flat blocks of steel, and tested to failure. Failure is defined as the first sign of a crack in the button visible to the naked eye or at the first audible sound of cracking. The load in pounds which produces this failure shall be recorded as the compressive strength of the button. Extreme care shall be taken to apply the pressure centrally and evenly over the entire surface of the button which is in contact with the steel blocks. When testing staple buttons, the shank shall be removed even with the back of the buttons. In case of conflict, test equipment with an automatic stress-strain recorder shall be used for referee testing. The rate of head travel shall not exceed

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0.050-inches per minute. The compressive strength shall be taken from the chart as the highest reading in pounds prior to the first drop in the curve which occurs when the button cracks or from a needle on the gauge indicating the maximum compression applied. The test may be discontinued when the load reaches 150 percent (%) of the required load without cracking occurring.

3.3.4 Resistance to water (Class 2 buttons only). Class 2 buttons shall show no crazing or distortion and shall not exhibit yellowing or loss of color or luster after being immersed in boiling distilled water for 1 hour and immediately thereafter dried and examined under the viewing conditions specified in 3.2.1. Any evidence of crazing, distortion, yellowing or loss of color or luster shall be considered a test failure.

3.3.5 Resistance to heat (Class 2 buttons only). Buttons shall show no softening, distortion, crazing, yellowing or loss of color when tested as follows: Class 2 buttons shall be sewn to cloth for testing. The sole temperature of a household flat iron shall be confirmed to be 275°F to 300°F with a surface pyrometer. The iron shall be applied to the button to be tested for one (1) minute. The button shall be examined under the viewing conditions specified in 3.2.1. Any softening, distortion, crazing, yellowing or loss of color or luster shall be considered a test failure.

3.3.6 Shank pull strength (Type II buttons only). Type II buttons shall have an average pull strength of not less than 20 pounds when tested with toggles or rings using accepted industry standard procedures. The results for all of the buttons in the sample shall be averaged and reported to the nearest 0.1 pound. Failure of the average to conform to the specified requirement shall be considered a test failure.

3.3.7 Salt-spray resistance (metal components). All metal parts, including toggles and rings (as applicable) shall be capable of withstanding a 96-hour continuous exposure to salt spray when tested in accordance with ASTM B117. Any rusting or corrosion products which cannot be removed when wiped with a soft cloth shall be cause for failure of the test.

3.4 Construction.

3.4.1 Type I design. Type I buttons shall be produced and finished to conform to the design shown in Figure 1 for the class, style and size specified (see 7.4).

3.4.2 Type II design. Type II buttons shall be produced and finished to conform to the design shown in Figure 2 for the class, style and size specified (see 7.4). The shank shall be firmly secured to the center back of the button and shall be supplied with a fastener device consisting of a metal button ring or metal toggle (see 3.1.2.1 and 3.1.2.2) conforming to the design and dimensions shown in Figure 2, unless otherwise specified.

3.5 Workmanship. The finished buttons shall conform to the quality of product established by this document. The occurrence of defects shall not exceed the contractor's own quality assurance standards and the quality assurance standards defined by the technical data and bid package.

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4. REGULATORY REQUIREMENTS. Unless otherwise specified, the offeror/contractor is encouraged to use recovered materials to the maximum extent practicable, in accordance with paragraph 23.403 of the Federal Acquisition Regulation (FAR).

5. PRODUCT CONFORMANCE PROVISIONS.

5.1 Product conformance. The buttons provided shall meet the salient characteristics of this CID; conform to the producer's own specifications, standards and quality assurance practices; and be the same product offered for sale in the commercial marketplace. The Government reserves the right to require proof of such conformance.

5.2 Market acceptability. The buttons, rings and toggles must have been sold to the commercial market or the Government.

5.3 End-item inspection procedures. Sampling for inspection shall be conducted in accordance with ANSI/ASQ Z1.4 – Sampling Procedures and Tables for Inspection of Attributes (see 7.2.2).

5.3.1 Visual examination of buttons, rings and toggles. Type I buttons and Type II buttons, toggles and rings shall be examined for the defects listed in Table III when viewed at a distance of 3 feet.

TABLE III. Visual defects – buttons, rings and toggles.

Examine	Defect
Buttons	
Workmanship	Nicked, gouged, cracked, scratched, dented, disfigured or malformed.
	Any fin, flash or chip
	Not clean - stained, spotted or inclusion of foreign matter
	Pitted, porous or rough surface
	Sharp edge or flash around the periphery or in a sewing hole (when in doubt, probe sewing hole with shank end of drill or needle of suitable diameter.
	Holes not uniformly spaced or any hole elongated
Shank (Type II button only)	Shank not firmly secured
	Shank bent or shank eye malformed
	Burr, fin, sliver, rough or sharp edge.
	Metal cracked or fractured.
Rings and toggles	
Workmanship	Burr, fin, sliver, rough or sharp edge

5.4 Acceptance criteria. Acceptance criteria shall be as specified in the contract or purchase order (see 7.5).

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6. PACKAGING

6.1 Packaging. Preservation, packing and marking shall be as specified in the contract or purchase order (see 7.5)

7. NOTES

7.1 Sources of Government documents.

7.1.1 Copies of Government documents are available online at https://quicksearch.dla.mil.

7.2 Sources of non-Government documents.

7.2.1 AATCC test methods are available for purchase online at https://www.aatcc.org.

7.2.2 ANSI/ASQ Z1.4 – Sampling Procedures are available online at https://www.asq.org.

7.2.3 ASTM test methods are available for purchase online at https://www.astm.org.

7.3 Intended use. Buttons covered by this CID are used primarily to secure clothing and equipment items, and may be used for decorative purposes.

7.4 Standard sample. For access to standard samples, address the contracting or ordering activity.

7.5 Ordering data. The contract or purchase order should specify the following:

a. Title, number, and date of this Commercial Item Description (CID)
b. Type(s), style(s), class(es) and size(s) required (see 2.1)
c. Color and finish required (see 3.2)
d. Product conformance provisions (see 5)
e. Packaging (see 6)

7.6 Changes from V-B-871F. Due to the extensiveness of the changes associated with converting from a specification to a CID, marginal notations are not used in this revision to identify changes with respect to previous issues. Type II, Class D and Type III, Class A have been redesignated as Type I, Class 1 and Type II, Class 1, respectively. Type II, Class K and Type III, Class E have been redesignated as Type I, Class 2 and Type II, Class 2, respectively. This change is reflected in Table IV below. In addition, Type I, Style 30 has been added to Figure 1 in this revision. This change is reflected in Table V below.

7.7 Historical changes. Some of the types, classes and styles of buttons covered by the historical specifications have been changed, discontinued and redesignated during previous revisions to V-B-871 as well as during the revision and conversion of V-B-871 from a Military Specification to a Commercial Item Description. See Tables IV and V for summarized changes. Italicized text references changes made between the latest version of V-B-871F and this CID.

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TABLE IV. Type and Class changes.

Type and Class in V-B-871	Current status
Type I, all Classes	Discontinued in V-B-871B
Type II	*Redesignated as Type I in this CID*
Class A	Discontinued in V-B-871B
Class B	Discontinued in V-B-871B
Class C	Discontinued in V-B-871B
Class D (1a)	*Active as Class 1 (redesignated in this revision)*
Class D (1b)	Discontinued in V-B-871B
Class E	Discontinued in V-B-871B
Class F	Discontinued in V-B-871B
Class G	Discontinued in V-B-871C
Class H	Discontinued in V-B-871C
Class I	Discontinued in V-B-871B
Class J	Discontinued in V-B-871B
Class K	*Active as Class 2 (redesignated in this CID)*
Type III	*Redesignated as Type II in this CID*
Class A (1a)	*Active as Class 1 (redesignated in this CID)*
Class B (1b)	Discontinued in V-B-871B
Class C	Discontinued in V-B-871B
Class D	Discontinued in V-B-871B
Class E	*Active as Class 2 (redesignated in this CID)*

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TABLE V. Style changes.

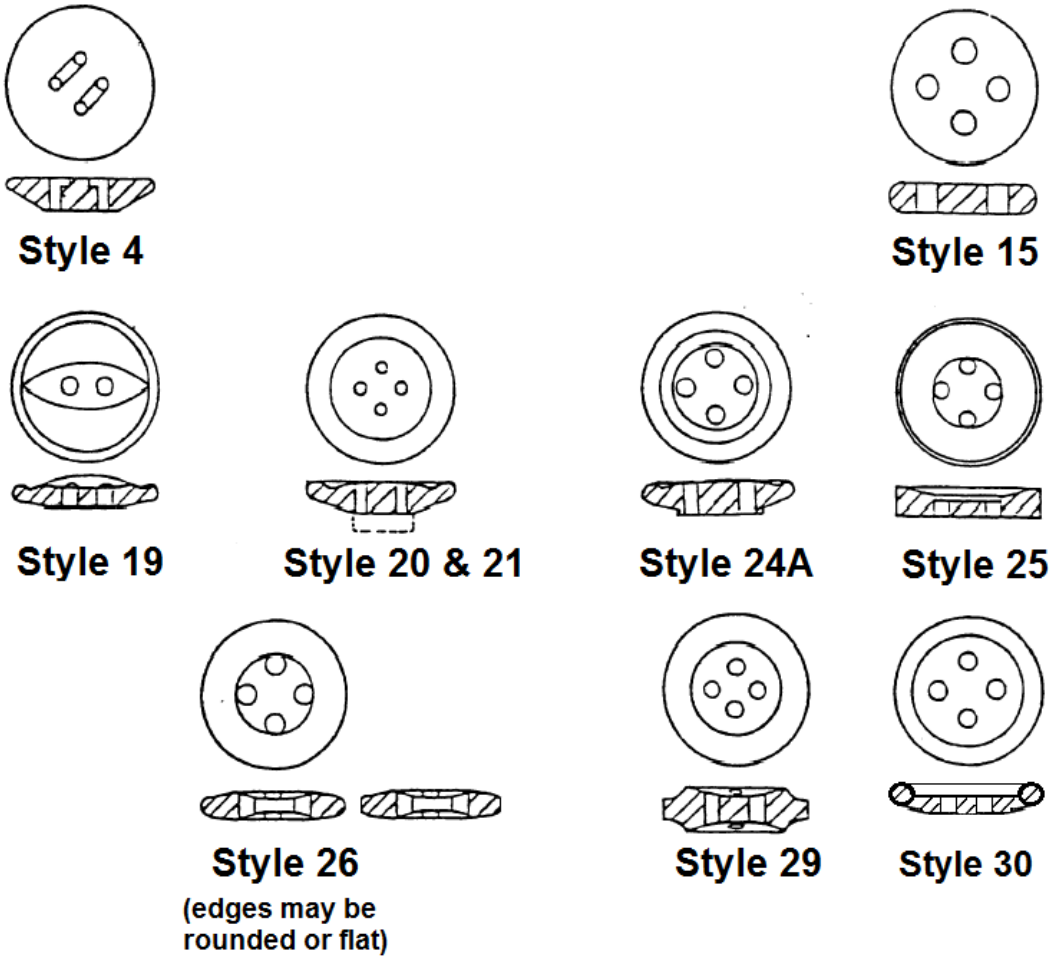
Type and Style in V-B-871	Current status
Type I, all Styles	Discontinued in V-B-871B
Type II	*Redesignated as Type I in this CID*
Style 1	Discontinued in V-B-871B
Style 2	Discontinued in V-B-871B
Style 3	Discontinued in V-B-871B
Style 4	Active as Style 4
Style 5	Discontinued in V-B-871B (Style 19 is suitable substitute)
Style 6	Discontinued in V-B-871B
Style 7	Discontinued in V-B-871B
Style 8	Discontinued in V-B-871C
Style 9	Discontinued in V-B-871C
Style 10	Discontinued in V-B-871B
Style 11	Discontinued in V-B-871B
Style 12	Discontinued in V-B-871B (Styles 20 and 21 are suitable substitutes)
Style 13	Discontinued in V-B-871B
Style 14	Discontinued in V-B-871C (Style 25 is suitable substitute)
Style 15	Active as Style 15
Style 16	Discontinued in V-B-871B
Style 17	Discontinued in V-B-871B
Style 18	Discontinued in V-B-871B
Style 19	Active as Style 19
Style 20	Active as Style 20 (also may use Style 21)
Style 21	Active as Style 21 (also may use Style 20)
Style 22	Discontinued in V-B-871B (Style 4 is suitable substitute)
Style 23	Discontinued in V-B-871B (Style 19 is suitable substitute)
Style 24	Active as Style 24
Style 25	Active as Style 25
Style 26	Active as Style 26
Style 27	Active as Style 27
Style 28	Discontinued in V-B-871C (Style 26 is suitable substitute)
Style 29	Active as Style 29
Style 30	*Active as Style 30 (added in this CID)*
Type III	*Redesignated as Type II in this CID*
Style 1	Active as Style 1
Style 2	Discontinued in V-B-871B
Style 3	Active as Style 4
Style 4	Discontinued in V-B-871B
Style 5	Active as Style 5
Style 5	Active as Style 6

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7.8 Subject term (key word) listing.

Formaldehyde
Ligne
Melamine
Shank
Toggle

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Note:

1. **Buttons of the same style but of different sizes may differ in appearance because of the changing ratios between button size, button thickness, sewing hole diameter and sewing hole spacing**
2. **Styles 19, 20 and 21. Stud sizes shall be as supplied by the manufacturer. Styles 20 and 21 may be supplied without studs.**
3. **Styles 1, 2, 3, 5 through 14, 16, 17, 18, 22, 23, and 28 have been discontinued. See Table V for more information.**

FIGURE 1. Type I button styles.

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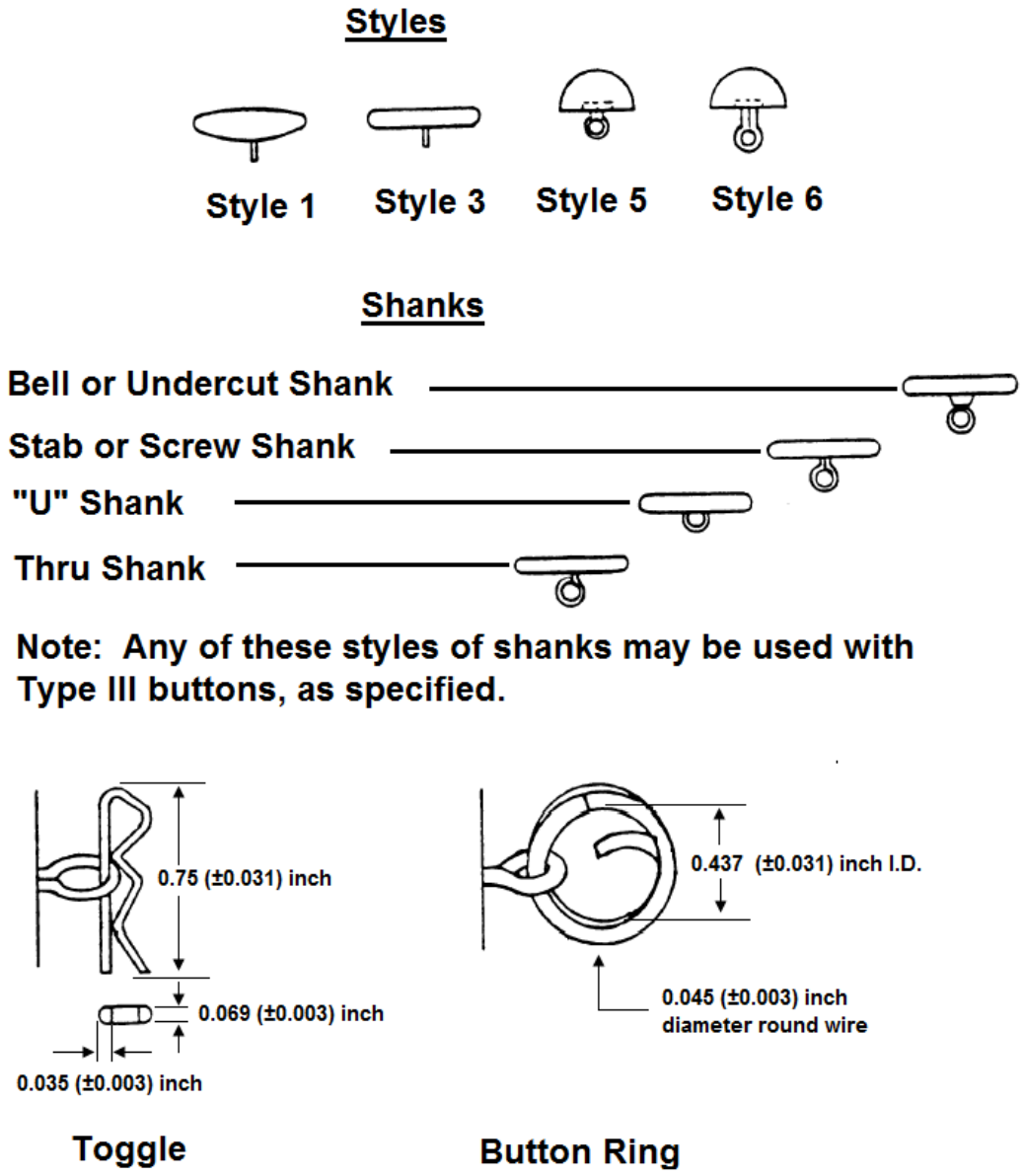


FIGURE 2. Type II button styles and toggle and ring description.

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MILITARY INTERESTS:

Custodian:
Army – GL
Navy – NU
Air Force – 11

Review Activities
Army – IH
Navy – MC

CIVIL AGENCY COORDINATING ACTIVITIES:
GSA - FAS

Preparing activity:
DLA - CT

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